

NORTHWEST ALABAMA, AL, USA

WMO: 723235

Lat: 34.744N

Lon: 87.600W

Elev: 165

StdP: 99.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.9	-4.8	-14.4	1.1	-3.9	-12.1	1.4	-2.4	10.6	12.1	9.1	12.6	2.8	50	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.6	24.1	34.3	24.0	33.2	23.8	26.1	32.0	25.5	31.4	25.0	30.7	3.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	19.9	28.8	23.9	19.2	28.2	23.4	18.5	27.6	81.5	32.1	79.0	31.4	76.7	30.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.6	-10.5	37.4	2.6	2.0	-12.4	38.9	-13.9	40.0	-15.3	41.2	-17.2	42.6		
(5)			-11.3	27.1	2.5	0.6	-13.1	27.6	-14.6	28.0	-16.0	28.3	-17.8	28.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	5.7	7.8	12.0	17.1	21.5	25.4	26.9	26.6	23.5	17.3	11.2	7.1	(6)	(7)
	DBStd	8.74	6.08	5.43	5.14	4.31	3.57	2.44	2.02	2.20	3.34	4.60	4.83	5.41	(7)	(8)
	HDD10.0	472	163	98	38	3	0	0	0	0	0	3	43	124	(8)	(9)
	HDD18.3	1629	393	296	204	73	14	0	0	0	4	76	219	348	(9)	(10)
	CDD10.0	2985	28	36	100	216	357	462	524	514	407	229	78	35	(10)	(11)
	CDD18.3	1102	1	2	8	36	113	212	266	256	161	43	4	1	(11)	(12)
	CDH23.3	10604	0	5	66	318	1014	2074	2678	2518	1474	426	30	2	(12)	(13)
(14) Wind	WSAvg	2.6	3.2	3.3	3.4	3.2	2.6	2.1	1.8	1.7	1.9	2.2	2.5	3.0	(14)	(15)
	PrecAvg	1378	115	115	149	112	128	115	113	85	104	78	116	143	(15)	(16)
	PrecMax	2001	299	273	362	325	301	341	366	182	257	210	281	487	(16)	(17)
	PrecMin	995	15	13	34	9	21	26	23	4	2	0	28	16	(17)	(18)
	PrecStd	278	59	59	79	65	65	66	69	49	60	56	52	91	(18)	(19)
	DB	21.9	24.0	27.9	30.3	33.4	36.0	36.9	37.9	35.9	31.7	26.6	23.1	19.3	(19)	(20)
	MCWB	16.9	17.4	18.1	19.6	22.4	23.8	24.3	24.0	22.3	21.0	18.9	18.3	16.9	(20)	(21)
(22) Mean Coincident Wet Bulb Temperatures	DB	19.5	21.6	25.2	28.5	31.7	34.2	35.2	35.6	34.1	29.5	23.5	20.7	17.6	(21)	(22)
	MCWB	15.6	16.2	16.9	18.9	21.9	23.7	24.9	24.3	22.6	20.6	16.8	15.3	13.2	(22)	(23)
	DB	17.6	19.3	23.0	26.9	30.3	33.1	33.9	34.0	32.3	27.7	21.6	18.3	15.3	(23)	(24)
	MCWB	14.7	14.5	15.7	18.2	21.2	23.5	24.7	24.0	22.6	19.4	16.1	15.3	13.2	(24)	(25)
	DB	15.5	17.0	20.8	25.1	28.8	31.8	32.6	32.5	30.4	25.8	19.7	16.1	13.2	(25)	(26)
	MCWB	12.2	12.8	14.7	17.4	20.7	23.1	24.4	23.9	22.1	18.4	15.1	13.2	10.0	(26)	(27)
	WB	18.3	19.0	19.7	21.7	24.3	26.2	27.1	26.9	25.1	23.5	20.5	19.3	17.6	(27)	(28)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	20.5	22.2	24.7	27.8	30.0	32.2	32.7	33.6	30.8	28.4	23.5	21.9	20.0	(28)	(29)
	WB	16.7	17.7	18.3	20.6	23.3	25.5	26.3	25.8	24.3	22.5	19.0	17.6	15.8	(29)	(30)
	MCDB	18.8	20.7	23.2	25.9	29.2	31.6	32.0	31.9	30.2	27.2	22.0	20.0	18.0	(30)	(31)
	WB	15.1	15.6	17.0	19.7	22.5	24.8	25.7	25.3	23.7	21.3	17.6	15.8	13.5	(31)	(32)
	MCDB	17.5	18.3	21.6	24.6	28.2	30.6	31.4	31.1	29.3	25.3	20.4	18.0	15.8	(32)	(33)
	WB	12.9	13.7	15.7	18.7	21.7	24.1	25.2	24.7	23.1	20.0	16.0	13.5	10.0	(33)	(34)
	MCDB	14.7	16.3	19.9	23.5	27.0	29.4	30.8	30.1	28.1	23.8	18.9	15.8	10.0	(34)	(35)
(35) Mean Daily Temperature Range	MDBR	10.1	10.6	11.5	12.1	11.5	11.2	10.7	11.4	12.3	13.0	12.2	10.0	8.4	(35)	(36)
	5% DB	11.3	13.1	14.1	13.8	13.3	13.3	12.5	13.7	14.2	15.1	14.6	11.7	8.4	(36)	(37)
	MCWBR	8.4	8.6	7.7	6.1	5.0	4.6	4.1	4.4	5.0	6.8	8.6	8.4	8.3	(37)	(38)
	5% WB	10.0	10.6	11.9	11.0	11.1	11.0	10.8	11.2	11.4	11.4	10.7	10.1	8.3	(38)	(39)
	MCWBR	8.4	8.1	7.5	5.7	4.9	4.7	4.2	4.3	4.6	6.0	7.6	8.3	8.3	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub	0.318	0.326	0.365	0.398	0.438	0.465	0.508	0.521	0.428	0.373	0.338	0.317	0.2514	(40)	(41)
	taud	2.494	2.470	2.386	2.288	2.223	2.171	2.080	2.031	2.282	2.416	2.457	2.514	2.514	(41)	(42)
	Ebn at Noon	889	921	906	888	853	826	788	769	831	854	855	868	868	(42)	(43)
	Edh at Noon	84	96	114	131	142	149	163	168	125	100	87	78	78	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg	2.45	2.97	4.12	5.30	5.87	6.28	5.99	5.54	4.81	3.83	2.79	2.09	0.21	(44)	(45)
	RadStd	0.20	0.37	0.37	0.36	0.45	0.49	0.33	0.31	0.42	0.50	0.29	0.21	0.21	(45)	(46)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-1.52	N/A	N/A	N/A	N/A	+146	N/A
(47)	Regional (28 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+140	+101

Nomenclature: See separate page